Technical review

Item #4 · screenshot.png · Run ID: 20261113T232757Z-cbb6f3

Path	`R:\evidence\images\screenshot.png`
Size (bytes)	193322
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.png
Detected type (signature)	PNG image
Status	Needs attention
Case	RV-SAMPLE-001
Evidence ID	E0014
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`ac7b0609981440312d29ab1926020de1d4889192b68d48f5b4960dc672e6bd47`
md5	`99a5e4d0099230fbc2521b5e82f1216c`

Findings

[Medium] Metadata dates do not agree (METADATA_DATES_INCONSISTENT)

What it shows	PNG tIME 2026-03-04 08:00:00 (UTC) and EXIF DateTime 2026:03:01 09:15:00 are 2.9 days apart.
What it does not prove	Clocks, time zones and software that rewrites dates all cause this; it does not show when the picture was taken or changed.

Image analysis

Status	Completed
Format	PNG
Dimensions (pixels)	400 × 300

Original image (decoded preview)

Lossless PNG of the decoded pixels, exactly as the analyses saw them (stored orientation; EXIF rotation not applied). The maps below have the same size and can be laid over it.



maps/original.png (400x300) — reduced copy for print; the full-resolution file is in the item folder

Metadata summary

Key fields only; metadata.json in the item folder holds every tag read. Metadata is written by the device or program that saved the file and can be edited or forged.

Tags read	8
Tags per group	Exif 3, IFD0 2, PNG-tEXt 1, PNG-tIME 1, XMP 1
GPS tags	absent
MakerNote	absent
XMP packet	present
Embedded thumbnail	absent
ICC profile (SHA-256)	—

Group	Field	Value
IFD0	Software	Screenshot tool (synthetic)
PNG-tEXt	Software	Screenshot tool (synthetic)
IFD0	DateTime	2026:03:01 09:15:00
Exif	DateTimeOriginal	2026:03:01 09:15:00
Exif	PixelXDimension	400
Exif	PixelYDimension	300
PNG-tIME	tIME	2026-03-04 08:00:00

Structure summary

Top-level segments in file order (JPEG markers, PNG chunks, RIFF chunks or HEIF boxes); structure.json holds the full tree with byte offsets.

Top-level segments	9
bit depth	8
colour type	2
bytes after IEND	0

Marker / chunk / box sequence:

`signature · IHDR · pHYs · eXIf · iTXt · tEXt · tIME · IDAT · IEND`

Fingerprint databases

Role	Database	Version	Entries	Status	SHA-256
starter	rakvor-starter	0	0	loaded	e0b0c87e4b79e5c5402e920d343436056c25bb959a5bb757a7d0519b95ab4be8
local	—	—	—	absent	—

Analysis maps

No analysis map was written for this item.

Copy-move shift vectors

No copy-move candidate was found (scaled or rotated copies are not detected).

Extracted files

structure.json (structure, 1739 B)	md5: 367a2eaef5ea949926d585e01d0254a1 sha256: 9bbeb201e1bff821bf6211c05c647e462d4a6d09ba231ccb18037ebc93929cd1
metadata.json (metadata, 1387 B)	md5: a9a88ba43f016cdac7bed1008b1b487c sha256: 0963f22ad40c840cc513ce4e86233b96796128273f70dcf5d8267efa84eed7a2
maps/original.png (original_preview, 205956 B)	md5: 338a961355fff7465260e891caa14b74 sha256: 2bfe41e3ea9b56619bff97b96cc02615b3b3f08bcb6324b15d51bb3efbb8036f

Method

Each file was opened read-only and hashed in a single pass with the selected algorithms, and its real type was identified from its signature. JPEG, HEIC, PNG and WebP images were then re-read in an isolated process and the bytes read were hashed again and compared with the first pass. The image analysis read the file structure and metadata, compared the JPEG quantization tables with the IJG standard and the fingerprint databases, looked for double JPEG compression, wrote the selected visual maps (error level, noise, JPEG ghosts), searched for copied-and-moved regions, compared the embedded thumbnail with the main image and, when reference images were given, compared the image with them. HEIC pixels are not decoded (structure and metadata only). Every action and every file written was recorded in a hash-chained audit log.

Audit trail

Audit entries	#32 – #39
Log folder	`R:\case\runs\rv-imgauth\20261113T232757Z-cbb6f3\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- An image without findings is not proven authentic: careful edits can leave none of the traces these checks look for, and it is impossible to prove a negative (SWGDE 18-I-001).
- Error level, noise and ghost maps are visual aids for a trained examiner; the tool gives them no score and draws no conclusion from them.
- Double JPEG compression shows the image was saved more than once; ordinary sharing and exporting do this.
- The compression-table database covers few devices and apps (built from Rakvor's own captures plus local learning); an absent or matching entry is weak evidence on its own.
- Copy-move detection compares blocks at the same scale and orientation on a downscaled image: scaled, rotated or small copies are not found.
- Error rates of these checks have not yet been measured on a documented validation corpus; the findings are signals to be weighed by the examiner.
- Modification times come from the file system of the examined copy and may differ from the original source.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.
- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.